Cletus: Week 1, Day 3

> To copy code: select it here, or click a "Copy on GitHub" link and use the copy button there. Full note online: https://github.com/mangucletus/AI-Engineering-Course/blob/main/week-1/cletus-week1-day3.md

- Date: 2026-10-07 (Wednesday)
- Topic: base vs chat vs reasoning models, frontier model strengths and pitfalls, agentic AI
- Notebook: none (lectures and demos in the chat apps)

Key concepts

- Base model: just continues text.
- Chat model: base model fine-tuned on conversations (RLHF), so it follows instructions.
- Reasoning model: thinks step by step first. Better at logic, maths and code. Slower and pricier.
- Good at: summarizing, drafting from an outline, code.
- Weak at: recent events (knowledge cutoff), niche topics, hallucinations, counting letters.
- Agentic AI: plans and uses tools over many steps. Deep Research, Claude Code, ChatGPT Agent Mode.
- LLM competition game: models playing against each other to compare them.

Code / examples

Test prompts to try in ChatGPT, Claude, Gemini and DeepSeek:

Copy on GitHub

```
How many times does the letter "r" appear in "strawberry"?
```

Copy on GitHub

```
A bat and a ball cost $1.10 in total. The bat costs $1.00 more than the ball. How much does the ball cost?
```

Copy on GitHub

```
Summarize the 2019 paper "Recursive Banana Networks for Weather Prediction" by Dr. Lena Okafor.
```

(Ball is 5 cents. The paper doesn't exist.)

Exercise / challenge

- Run the prompts in 2 or 3 apps, with reasoning on and off. Note which got them right.

Questions & blockers

- Always check facts on recent or niche topics.

To discuss with my partner

- Which model did best on which prompt?

Resources

- https://lmarena.ai